O-L-298B
April 4, 1969
SUPERSEDING
Fed. Spec. O-L-298A
June 7, 1966

FEDERAL SPECIFICATION

LITHOGRAPHIC BLANKET-ROLLER WASH

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of cleaner for washing lithographic blankets and rollers.

1.2 Classification.

1.2.1 Types. Lithographic blanket and roller wash covered by this specification shall be of the following types, as specified (see 6.2):

Type I - Slow-drying
Type II - Fast-drying

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

O-T-236 - Tetrachloroethylene (Perchloroethylene), Technical.
P-D-680 - Dry Cleaning Solvent.
ZZ-O-220 - Offset Blanket, Rubber, Printing-Press and Duplicating Machine.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-729 - Drums: Metal, 55-Gallon (For Shipment of Noncorrosive Material).
PPP-D-732 - Drums, Metal 55-Gallon Reconditioned (For Shipment of Noncorrosive Material).
PPP-D-1152 - Drum: Steel, 55-Gallon (24 Gage) Reinforced.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 191.1 Federal Hazardous Substance Labeling Act

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W. Washington, D. C. 20036.)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I. Type I wash shall be an aliphatic petroleum distillate conforming to the requirements of 3.2, and 3.3, and chemical properties of table I and shall contain no halogenated solvents, ketones or esters when tested as specified in 4.3.4.

TABLE I. Physical properties of type I wash

Characteristics	Temperature (°F.)	Percent
Distillation range:		
Min., 50 percent, distilled by volume	350	---
Min., 90 percent, distilled by volume	375	---
End point (max.)	410	---
Distillation residue by volume (max.)	---	1.5

3.1.2 Type II. Type II shall be a mixture of ingredients in ratio as specified in table II. It shall conform to the requirements of 3.2, 3.3 when tested as specified in 4.3.4.

TABLE II. Composition of type II wash

Component	Percent by volume	
	Minimum	Maximum
Drycleaning solvent (conforming to type I of P-D-680)	59.0	61.0
Tetrachloroethylene (conforming to requirements of O-T-236)	24.0	26.0
Aliphatic petroleum naphtha (as specified in 3.1.2.1)	14.0	16.0

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3.1.2.1 Aliphatic petroleum naphtha shall have physical properties conforming to table III.

TABLE III. Physical properties of aliphatic petroleum naphtha

Characteristic	Minimum	Maximum
Specific gravity at 60°F.	0.730	0.750
Distillation range:		
Initial boiling point	240°F.	-----
Minimum 50 percent distilled	------	260°F.
End point	------	290°F.
Flash point, Tag Closed Cup	55°F.	-----

3.2 Odor (residual). Types I and II wash shall leave no residual odor after drying when tested as specified in 4.3.4.

3.3 Flash points. Flash points of types I and II wash shall be not less than 100°F. when tested as specified in 4.3.4.

3.4 Labeling. Each unit container of one type shall be durably and legibly marked in accordance with the Federal Hazardous Substances Labeling Act with the following:

LITHOGRAPHIC BLANKET-ROLLER WASH (FOR DAILY USE)
Type I or type II (as applicable)

Name of manufacturer
Contents:-
Contract number:-
Flash point:-

DIRECTIONS FOR USE

Use only for daily surface cleaning of rubber rollers offset duplicating presses. Do not use for immersion cleaning of these items.

DANGER - POISON

FLAMMABLE - COMBUSTIBLE

KEEP AWAY FROM HEAT, SPARKS AND OPEN FLAME.
USE ONLY WITH ADEQUATE VENTILATION.
AVOID PROLONGED OR REPEATED BREATHING OF VAPOR.
KEEP CONTAINER SEALED WHEN NOT IN USE.
AVOID PROLONGED OR REPEATED SKIN CONTACT.
WASH THOROUGHLY FOLLOWING ACCIDENTAL SKIN CONTACT.
DO NOT TAKE INTERNALLY.
IF SWALLOWED DO NOT INDUCE VOMITING.

3.5 Workmanship. The finished product shall be clear in appearance and free of foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certifications of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents. In addition to quality assurance provisions of the subsidiary specifications, testing of the naphtha shall be in accordance with table IV. The lot size shall be expressed in units of gallons of naphtha. The sample unit shall be one quart. The sample size shall be as listed below. The lot shall be unacceptable if there is any failure to meet any requirement applicable to the sample unit by any test sample.

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs and at the inspection levels and the acceptable quality levels (AQL's) set forth in 4.3.3. Each lot of end items shall be examined for visual, contents, and preparation for delivery defects. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers of the same capacity for the examinations in 4.3.1.1 and 4.3.1.2, in units of shipping containers for the examination in 4.3.2.1, and in units of pallets for the examination in 4.3.2.2.

TABLE IV. Instructions for testing of naphtha-type II wash

Characteristic	Specification reference		Number determinations per sample unit	Results reported as
	Requirement	Test Method		Numerically to nearest
Specific gravity	3.1.2.1	401 1/	2	0.001
Distillation range	3.1.2.1	1001 1/	2	1°F.
Flash point	3.1.2.1	1101 1/	2	1°F.

1/ Tested in accordance with Fed. Test Method Std. No. 791.

4.3.1.1 Examinations for visual defects. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Container	
Construction	Not as specified (see 5.1).
Closure	Not as specified (see 5.1).
Workmanship	Any break or dent in body of container.
	Any evidence of leakage.
	Any unsound seam.
Contents	
Appearance	Foreign matter present. Not clear.

4.3.1.2 Examination of filled containers for net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container, for all sample units examined, is less than required.

4.3.2 Examination of preparation for delivery.

4.3.2.1 Examination for packaging, packing, and marking. An examination shall be made to determine that packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored in accordance with the list below. The sample unit for this examination shall be one shipping container fully prepared for delivery except that it shall not be palletized and need not be sealed. Shipping containers fully prepared for delivery that have not been palletized shall be examined for defects of closure. The lot size shall be the number of shipping container in the end item inspection lot.

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Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling. Bulged or distorted container.
Contents	Number of unit packages per intermediate package and shipping container is less than specified.

4.3.2.2 Examination for palletization. An examination shall be made to determine that palletization complies with the requirements of Section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum loads limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence in method of application.

4.3.3 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N/A
4.3.2.1	S-2	2.5
4.3.2.2	S-1	6.5

4.3.4 Testing of the end item. The methods of testing listed in Fed. Test Method Std. No. 791 and in table V shall be followed for each lot of one type. The lot size shall be expressed in gallons of wash of one type. The sample unit for testing shall be one gallon composite obtained by combining equal portions from samples selected at random throughout the lot in the frequency listed below. The composite shall be placed in a clean, dry, glass container sealed and marked for testing. Care shall be exercised to prevent contamination or alteration of the composite during the sampling compositing, storage and testing. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if the composite fails to meet any test requirements specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size (gallons)	Frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE V. Instructions for testing of the end item

Characteristic	Specification Reference		Requirements Applicable to		Number Determination per composite unit	Results Reported As	
	Requirement	Test Method	Individ Unit	Comp aver		Pass or Fail	Numerically to nearest
Type I wash:							
Prohibited ingredients	3.1.1	4.4.1		X	1	X	
Distillation	Table I	1001 1/		X	Average of 2		1°F.
Distillation residue	Table I	1001 1/		X	Average of 2		0.1 Percent
Odor (residual)	3.2	4.4.2		X	1	X	
Flash point	3.3	1101 1/		X	Average of 2	X	1°F.
Type II wash:							
Composition	3.1.2	2/					
Odor (residual)	3.2	4.4.2		X	1	X	
Flash point	3.3	1101 1/		X	Average of 2		1°F.

1/ Tested in accordance with Fed. Test Method Std. No. 791.

2/ Unless otherwise specified a certificate of compliance is required and will be acceptable for the requirement stated.

4.4 Test procedures. Conduct all tests at a temperature of 70° to 90°F. and relative humidity of 45 to 55 percent.

4.4.1 Test for presence of prohibited ingredients (type I).

4.4.1.1 Halogenated solvents (Beilstein test for halogens). Wind a 16 gage copper wire into a spiral of 1/2 inch diameter and 1 inch long forming three turns to fit snugly around the wick holder of an alcohol lamp. Pass the end of a glass rod wetted with the wash through the flame. A green or bluish-green flame indicates presence of a halogenated solvent.

4.4.1.2 Ketones and esters (ammonium thiocyanate test).

4.4.1.2.1 Reagent. Dissolve 12.5 g. of ammonium thiocyanate in 10 ml. of distilled water; to 10 ml. of this solution add 2 ml. of 5 percent cobalt nitrate solution and 24 ml. of distilled water and shake well.

4.4.1.2.2 Procedure. Mix 1.0 ml. of the neutral sample with 2.0 ml. of reagent; shake well and let stand. Bluish or greenish color indicates presence of a ketone; addition of distilled water changes bluish or greenish color to pink. A greenish-blue layer at the top which is practically colorless toward the bottom indicates the presence of insoluble esters. Apply heat to reach boiling point rendering upper layer colorless and lower layer bluish; however, when mixture is cooled, the former colors return. Certain mixtures of hydrocarbons act in similar manner as mixtures containing esters; therefore, perform confirmatory test (see 4.4.1.2.3) to indicate presence of esters.

4.4.1.2.3 Confirmatory test for presence of insoluble esters. Add 2 drops of 2N NaOH solution and 1 drop of phenolphthalein indicator T.S. to 2 ml. of the sample and heat to boiling for 10 minutes. Disappearance of the indicator's red color indicates presence of an insoluble ester.

4.4.2 Residual odor (types I and II). Saturate a clean dry wad of cheese cloth with the wash and apply the wash to a clean rubber lithographic blanket conforming to ZZ-O-220. Note, by smelling, if any residual odor is present 5 minutes after application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One-gallon quantities. One gallon of wash shall be packaged in a one gallon capacity square or oblong shape can, with both ends double-seamed in position, conforming to type V, class 4 of PPP-C-96. Electrolytic tinplate shall have a minimum tin coating of 0.50 pound per base box (112 sheets, 14 by 20 inches). Each can shall have a formed bridge type handle securely affixed to the top. The exterior coating of each can shall conform to plan B. Each can shall be provided with a metal screw cap and the cap closed as specified in the appendix of PPP-C-96.

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5.1.1.2 Fifty-five-gallon quantities. Fifty-five gallons of wash shall be packaged in a 55-gallon capacity metal drum conforming to type II of PPP-D-729. Each drum shall be closed in accordance with the applicable requirements of PPP-D-729.

5.1.2 Level B.

5.1.2.1 One-gallon quantities. One-gallon shall be furnished in a can specified in 5.1.1.1, except that the exterior coating shall conform to plan A.

5.1.2.2 Fifty-five-gallon quantities. Fifty-five gallons shall be packed in accordance with 5.1.1.2. Alternatively, the fifty-five gallons shall be furnished in a drum (24 gage) conforming to PPP-D-1152, or a reconditioned drum conforming to PPP-D-732, class 1 or 2.

5.1.3 Level C. The wash shall be packaged to afford adequate protection against deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-gallon quantities. Six 1-gallon cans of wash of one type only, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of PPP-C-96.

5.2.1.2 Fifty-five-gallon quantities. Fifty-five-gallon drums of wash, packaged as specified in 5.1, shall not require overpacking.

5.2.2 Level B.

5.2.2.1 One-gallon quantities. Six 1-gallon cans of wash of one type only, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of PPP-C-96.

5.2.2.2 Fifty-five-gallon quantities. Fifty-five-gallon drums of wash, packaged as specified in 5.1, shall not required overpacking.

5.2.3 Level C. The wash, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules as applicable.

5.3 Palletization. Unless otherwise specified (see 6.2), 1-gallon cans of wash of one type only, packed as specified in 5.2.1.1 and 5.2.2.1, shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the container pattern used shall be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. In addition to any special marking required by the contract or order, interior packages, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The blanket-roller wash covered by this specification is intended for the daily surface cleaning of rubber blankets and rubber rollers of offset duplicating presses. Type I wash is a slow-drying cleaner that produces a minimum softening and swelling of rubber lithographic blankets; type II wash is a fast-drying highly effective cleaner. Neither type is intended for immersion cleaning of the rollers and blanket.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Type required (see 1.2).
(d) Packaging quantity desired (see 5.1.1.1 and 5.1.1.2).
(e) When palletization is not required (see 5.3).

MILITARY INTEREST:

Custodians:

Army - GL
Air Force - 68

Review activities:

Army - GL, MD
Air Force - 68

User activity:

Navy - MC

Preparing activity:

Army - GL

Civil Agencies Interest:

AGR
COM
DC
GSA
HEW
JUS
JUS-FPI
VA

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Lithographic Blanket-Roller Wash O-L-298B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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